

Low Offset and Noise in High Biased GaN 2DEG Hall-effect Plates Investigated with Infrared Microscopy

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Abstract—This paper presents GaN two-dimensional electron gas (2DEG) Hall plates with low residual offset and noise at 3 V input bias. We studied devices made from three consecutive fabrication generations through current spinning offset measurements in a zero-field chamber. When operated above 1 V, the first-generation devices charted high residual offsets > 1 mT. We reduced these residual offsets by three orders of magnitude in later device generations. Two experiments were performed to confirm the improvements. First, the GaN 2DEG Hall-effect plates were measured with infrared microscopy during current spinning to investigate current crowding and Joule heating. The low offset devices had minimal thermal gradients while the high offset devices had large thermal variations during current spinning. Second, zero-field current spinning was used to compare GaN 2DEG Hall-effect plates against typical commercial silicon Hall-effect plates. The Si devices had residual offsets > 10 μ T biased at 3 V and AlGaIn/GaN devices had residual offset < 4 μ T at 3 V, and an InAlN/GaN devices operated at 2 V had a residual offset < 4 μ T. Thus, for the first time, GaN 2DEG Hall-effect plates were shown to operate with high bias with low offsets and noise, which enables high fidelity sensing applications in extreme environments.

Index Terms—Hall-effect, 2DEG, offset, current spinning, GaN, infrared, noise

I. INTRODUCTION

Hall-effect plates are advantageous for magnetic field sensing in a wide variety of consumer, automotive and industrial applications because of their low cost and ease of integration with CMOS electronics. To extend the scope of Hall-effect devices to harsher environments, they need to be able to withstand wide temperature ranges, operate with high sensitivity, and have low noise floor and DC offset. Though silicon (Si) Hall-effect plates have been shown to operate from

-50°C to 200°C with temperature stability on the order of 300 ppm/K [1], their performance is limited outside of this temperature range [2]. In addition, Si Hall-effect plates have noise floor limitations because they are low-doped and have $k\Omega$ range impedance [3-5].

Hall-effect plates based on heterostructures with a two-dimensional electron gas (2DEG) have gained popularity in recent years due to their high sensitivity and small residual offsets [6-8]. In particular, GaN-based Hall-effect devices can operate in extreme environments that are inaccessible to lower bandgap materials like Si [9]. For example, we have recently shown the operation of GaN 2DEG devices at 600°C for over 12 hours with high thermal stability [9]. Future monolithic integration platforms are also being developed in GaN devices for higher temperature operation [10]. While current spinning methods have been implemented in Si and GaAs Hall-effect plates to achieve residual offsets ranging from 0.1 to 1 μ T [6-8,11,12], our previous work showed that GaN device operation with low residual offsets was limited to small voltages (< 1 V) [8,13]. In order to maintain high signal-to-noise ratio with high bias operation, this residual offset needs to be reduced. Unlike their Si MagFET predecessor [14], the 2DEG channel is “normally-on”, meaning it does not require inversion from a field plate potential. This is beneficial because it will not suffer conduction pinch-off at high bias operation, nor body bias conduction modulation of typical Si Hall devices [12].

In this work, we confirmed our previous hypothesis that the high residual offsets at > 1 V operation in our first GaN Hall-effect plates were predominantly from self-heating effects with poor contacts [13]. We recorded the hot spots throughout a current spinning cycle with an infrared (IR) microscope. In addition, we document the improvements made through three generations of GaN 2DEG Hall-effect plates to surpass typical commercial Si Hall-effect plates in terms of reduced noise and

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